

Select → From → Where → group by → having → order by

Select productdesc *think if, elif, else*
Case when productdesc like '.i.desc' then 'desk'
when productdesc like '.i.table' then 'table'
else 'something else'
End AS short_desc
From product_T;

Functions: avg(col)
max(col), sum(col),
Count(col), lower(col)
nvl(null, 1) → changes null values

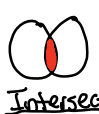
Subqueries → executed first!!!
noncor → do not use outer
every data, once for entire outer
Corr → use outer every data,
once for each row (uses exists)
*NEVER = Null, must be "Is Null" or "Is Not Null"

Where boolean ops → Not, and, or use parentheses()

Ranges → between (inclusive) otherwise use >= ops, can also do NOT between



Union



Intersect



Minus

Select column_names from table_1
Union (or any other) All → allows duplicates
Select column_names from table_2

Joins: Inner Join, Natural Join (removes on clause, auto match)
Left or Right [outer] Join, Full [outer] Join
Select columns From table_1
Right Outer Join table_2
On table1.column = table2.column;
Note: Self Joins use aliases declared in from clause
In and Not In:
Select ...
... where custstate in ('NJ', 'NY');
or
Not in ('pa', 'ca');

Having → after grouping
*Can use sum, count etc...
Select productID, Count(*)
From product_T
Group by productID
Having Count(*) > 2;

Group by → "Buckets"

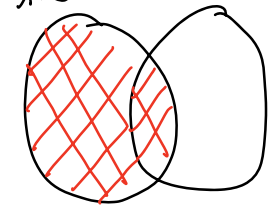
Select prodlineID, Count(*)
From product_T
Group by prodlineID
*Count how many prods in each prod line

Right and Left Joins prioritize that tables data!!!

*Trace through merge statements to understand what they do + how they do it.

*Joins used for different tables

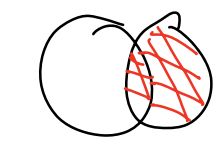
Multi table Joins work off the table created in first join
Select '5+7' From dual; → for math and printing



→ Left Join

Like '%Ben' → ends in Ben
Like '%. Ben%' → contains Ben

case sensitive *MERGE

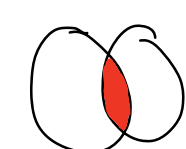


→ Right Join

Merge #Cust AS Target → target table changes
Using #custog AS Source
ON target.custnum = source.custnum
When Matched ...
Then
update set ...
When Not matched by target
then
insert ...
When Not matched by source then delete
think if, elif, else + aliases



→ Full Join



→ Inner Join

*set operations same table or similar data

Group by = Groups; Where = Rows

HINTS Use distinct or group by when case
Look for '...' → string, Minus ex: {1,2,3} - {2,4} = {1,3}

NULL is never =
Group by needs aggregates
Multi table Joins work off RESULTS
Null is "null" not empty
From products left join inventory (keeps prods)
From inventory right join products (keeps inventory)